26900059 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBTG301 - ENZYME TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Which enzyme helps digest starch? [K] / [CO1]
2. What are biological catalysts called? [K] / [CO1]
3. What happens to an enzyme's structure when it loses its functional conformation? [U] / [CO1]
4. Recall the EC class involved in molecular rearrangements. [K] / [CO1]
5. What type of inhibitor competes with the substrate for the active site? [K] / [CO2]
6. Connect enzyme stability with enzyme shelf life. [An] / [CO4]
7. Name the two techniques in which an enzyme is respectively attached to a surface and trapped inside a gel. [K] / [CO4]
8. Explain the role of GGT in the interpretation of increased ALP. [U] / [CO4]
9. Name two major sources of industrial enzymes. [K] / [CO4]
10. What are PETases? [U] / [CO4]

11. Analyse the relationship between enzyme specificity and the efficiency of enzymatic bioremediation. [An] / [CO4]

12. Which phenomenon causes reduced reaction rates inside immobilized enzyme matrices? [U] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Compare absolute specificity, group specificity, and stereospecificity with suitable examples. [U] / [CO1]

14. Classify fumarase based on its catalytic activity and explain the type of reaction it performs. [U] / [CO1]

15. Connect improved catalytic activity of engineered enzymes with increased industrial productivity [An] / [CO4]

16. Describe how binding of a substrate at one active site can affect the other active sites of an allosteric enzyme. [U] / [CO3]

17. Describe the effect of temperature on enzyme activity. [U] / [CO2]

18. Explain the role of synthetic enzymes in biosensors. [U] / [CO4]

[5x4 = 20]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

19. Explain cofactors and their role in enzyme catalysis. [U] / [CO1]

20. Explain the Michaelis–Menten equation in detail and discuss the significance of Km and Vmax in enzyme kinetics. [U] / [CO2]

[10x1 = 10]